4 Fig. 4 shows the regular octagon ABCDEFGH.

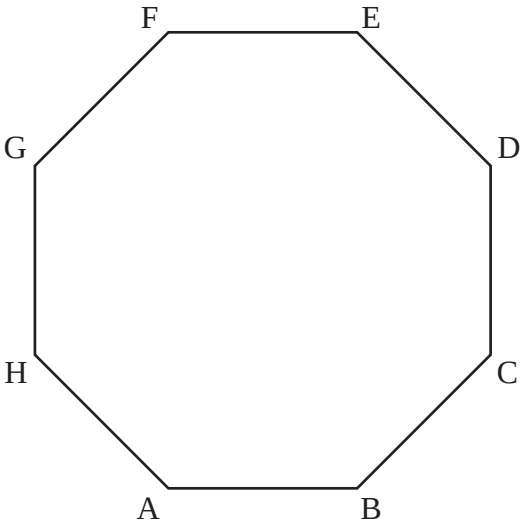

Fig. 4

$\overrightarrow{AB} = \mathbf{i}$, $\overrightarrow{CD} = \mathbf{j}$, where **i** is a unit vector parallel to the x-axis and **j** is a unit vector parallel to the y-axis.

Find an exact expression for $\overrightarrow{BC}$ in terms of **i** and **j**. **[3]**